

Session 2 — First night: eyes only

Practical Astronomy Intensive

9 questions 20 minutes 8 concepts 7 diagnostic

The first observing night, checked partly at the eyepiece-less sky itself: two of these are practical tasks, because averted vision and tracing the Milky Way are things you watch a student do, not things you read about their doing. Dark adaptation is asked twice — it is the night's discipline, and one white phone screen undoes it.

- 1 Standing outside, you can see steely-blue Vega, golden Arcturus and red Betelgeuse without any instrument. What does a star's colour tell you?

stars-vary Apply Multiple choice

- a Its surface temperature, directly — blue Vega is genuinely hotter than red Betelgeuse ✓
- b Nothing real — all stars are alike, and colour is just an effect of how far away they are
REVEALS Stars believed identical, differences attributed to distance alone
- c Its temperature, with red the hottest and blue the coolest
REVEALS Hot-tap habit imported to stars — red read as hot, blue as cold
- d How near it is — the reddest stars are the closest, which is why their colour shows
REVEALS Colour or brightness read as a distance indicator

Colour is a direct read-out of surface temperature, and the scale runs opposite to the bathroom tap: blue is hot, red is cool. Brightness is a different question again — it mixes true output with distance, which is why Betelgeuse blazes from over 500 light-years away.

- 2 A student looks at the Orion Nebula and complains that it is grey, nothing like the red and blue photograph he has seen. What is the honest explanation?

eye-at-night Apply Multiple choice

- a His eyes work in the dark just as in daylight, only dimmer, so he needs to look harder
REVEALS Night vision as a dimmed version of day vision; the cones' shutdown not understood
- b At that light level only his rods are firing, and rods are colour-blind — the photograph stacked photons for an hour, which no eye can do ✓
- c The telescope's optics are poor — a better instrument would show the colours
REVEALS Grey nebulae blamed on the telescope rather than on retinal physiology
- d He should stare directly at the nebula's centre, where his vision is sharpest, and the colour will emerge
REVEALS Direct vision believed best for faint objects — averted vision not understood

Cones need light and report colour; rods work at low light and report grey. Colour is the first thing to go in the dark. The centre of the retina is nearly all cones — the worst possible detector for something faint, which is why averted vision works.

- 3 Twenty-five minutes into the dark, a student pulls out his phone to check the time. The screen is white. What has he just done?

dark-adaptation

Apply

Multiple choice

- a Lost a little sensitivity, which will come back within a minute or two outside
REVEALS Dark adaptation believed to take a minute or two rather than 20–30
- b Nothing to his eyes — dark adaptation is about learning to concentrate, not about the retina
REVEALS Dark adaptation and averted vision treated as mental tricks rather than retinal chemistry and anatomy
- c Nothing much — one quick glance at a screen is harmless
REVEALS White light believed harmless in a single brief glance
- d Bleached his rods and handed back most of the twenty-five minutes — rhodopsin has to rebuild from scratch, which is exactly why torches are red ✓

Rhodopsin takes 20–30 minutes to rebuild and a white screen bleaches it in a second. This is chemistry, not attitude — hence the red torch rule, which needs no further explanation once the group has watched it happen once.

- 4 You are shown a faint object (M33, or a dim cluster member the assessor names). Find it and demonstrate averted vision, describing aloud what changes.

dark-adaptation

Apply

Practical task

ANSWER / WHAT TO WATCH FOR

Assessor watches for: student has been in the dark, red torch only, no white screens for the preceding 20+ minutes; does not immediately stare dead-on and give up; deliberately fixes their gaze 10–20° to one side of the target while attending to it; reports that the object brightens or appears when looked beside and fades when looked at directly. Credit the student who names the reason — the off-centre retina is packed with rods — rather than treating it as a trick. Do not credit a student who stares straight at it, declares nothing is there, and reaches for a brighter eyepiece.

Averted vision is plain retinal anatomy: the centre of the retina is nearly all cones, useless at this light level, and the off-centre retina is rod-rich. Watching a student actually do it is the only honest check — every one of them can already recite the definition.

- 5 Someone points at the Big Dipper (Saptarishi) and calls it a constellation. What is the correct picture?

asterisms

Recall

Multiple choice

- a It is an asterism, and since asterisms are unofficial they are of little real use for finding your way around
REVEALS Asterisms dismissed as second-rate because they are unofficial
- b It is an asterism — the bright core of Ursa Major, which is a far larger patch of sky — and asterisms are exactly what observers navigate by ✓
- c It is an asterism, meaning its seven stars are a genuine group close together in space
REVEALS Stars of an asterism believed to belong together physically
- d They are right — the Big Dipper is one of the 88 official constellations
REVEALS Big Dipper / Saptarishi believed to be a constellation rather than an asterism within Ursa Major

Asterisms are informal patterns: the Big Dipper is the bright core of the much larger Ursa Major, and the Summer Triangle's three stars lie in three different constellations at three very different distances. Unofficial does not mean useless — they are the sky's working landmarks.

6 A student's cousin has been given a certificate from a 'star registry' naming a star after her. What should you tell her?

star-names

Recall

Multiple choice

- a Only the IAU names stars — the certificate is a souvenir and that name appears in no catalogue any astronomer uses ✓
- b The name is official once registered, though it takes some years to appear in catalogues
REVEALS Belief that a star can be bought and officially named
- c It is fine as long as the star was an alpha star, since alpha is always the brightest of its constellation and therefore properly catalogued
REVEALS Alpha assumed always to be the brightest star of its constellation
- d It cannot be right, because each star has exactly one true name and that one is already taken
REVEALS Belief that a star has one single true name, rather than several designations from different systems

Star naming is the IAU's, and the registries sell paper. Note also that Bayer's lettering is rough — Betelgeuse is alpha Orionis but Rigel outshines it — and one star legitimately carries several names at once: Aldebaran, alpha Tauri and a catalogue number are the same star under systems built for different purposes.

7 A student holds his planisphere flat like a road map, facing south, and cannot make anything match. Later at the eyepiece he hops confidently in the wrong direction. What has gone wrong in each case?

star-charts

Reason

Multiple choice

- a A sky map only reads correctly when turned to face the direction you are looking, horizon edge down — and at the eyepiece a diagonal flips the view left for right while a Newtonian turns it round ✓
- b He should hold the chart fixed as he did; the eyepiece view matches the chart exactly, so the hop must have been mis-counted
REVEALS Chart held fixed like a road map; eyepiece view expected to match the chart's orientation
- c Nothing is wrong with the chart handling — the chart's stars are simply plotted for a different date
REVEALS Chart orientation and eyepiece flip not recognised as the problem at all
- d Both problems disappear with a planetarium app, which orients itself and makes paper charts unnecessary
REVEALS Planetarium app assumed to replace charts — white screen and lost patterns ignored

Two separate rotations bite beginners. A planisphere is turned to face your viewing direction with the horizon edge down. And the eyepiece image is not the chart image: a star diagonal mirror-reverses it, a Newtonian rotates it 180°. An app is not the cure — its white screen wrecks dark adaptation and zoomed in it hides the very patterns you navigate by.

- 8 A student in a Bortle 8 city backyard wants to buy a light-pollution filter so she can see galaxies from home. What is the honest advice?

light-pollution

Apply

Multiple choice

- a It will not work for galaxies — a broadband filter rejects a few lamp wavelengths and does nothing for broad-spectrum targets; that money is better spent on the drive to a dark site ✓
- b There is nothing to be done — a bright city sky is simply how cities are
REVEALS Skyglow accepted as inevitable rather than caused mostly by unshielded upward-throwing fixtures
- c It is worth it, though light pollution is really only an inconvenience for astronomers anyway
REVEALS Light pollution framed as an astronomers-only complaint
- d Buy it — a light-pollution filter effectively restores a Bortle 8 sky for any target
REVEALS Light-pollution filter believed to fix a bright sky, galaxies included

Filters reject narrow lines; a galaxy's light is broad starlight and gets blocked with the skyglow. Worth noting too that most skyglow comes from unshielded fixtures throwing light sideways and upward — it is fixable, it is on someone's electricity bill, and it disrupts nesting birds and human sleep as well as astronomy.

- 9 Trace the Milky Way band across the sky with your arm, name the constellation the brightest part runs through, then put binoculars on it and tell the group what you see.

milky-way-band

Apply

Practical task

ANSWER / WHAT TO WATCH FOR

Assessor watches for: student is dark-adapted before attempting this; traces the band as an arch across the whole sky rather than pointing at one patch; identifies the bright, crowded region toward Sagittarius (the galactic centre direction); on raising binoculars, reports that the 'cloud' breaks into countless individual stars. The claim to hear is that we are inside the disc and the band is simply the crowded direction along it. Probe once: 'so is the Milky Way that band over there?' — credit the student who answers that every naked-eye star, including the ones behind us, belongs to it.

Three wrong models die at once here: that the band is gas or cloud (binoculars resolve it into stars), that the Milky Way is an object off in the sky (we are inside it), and that those sweeping spiral pictures are photographs of it (nobody has ever been outside — every one is an artist's impression or another galaxy).